Case Statements Exercise 3

(1)

TABLE: PRODUCTS

PRODUCT_ID	PRODUCT_NAME	PRICE
1	LAPTOP	1200,00
2	PHONE	800,00
3	KEYBOARD	45,00
4	MONITOR	300,00
5	MOUSE	25,00

QUESTION: CLASSIFY EACH PRODUCT BY PRICE

- 'EXPENSIVE' IF PRICE > 1000,00
- 'MID-RANGE' IF PRICE BETWEEN 100 AND 1000,00
- 'BUDGET' IF PRICE < 100,00

EXPECTED OUTPUT COLUMN: PRODUCT_NAME
PRICE
PRICE_CATEGORY

```
SELECT PRODUCT_NAME, PRICE,
CASE
WHEN 'PRICE' > 1000,00 THEN 'EXPENSIVE'
WHEN 'PRICE' BETWEEN 100,00 AND 1000,00 THEN 'MID-RANGE'
WHEN 'PRICE' < 100,00 THEN 'BUDGET'
END AS 'PRICE_CATEGORY'
FROM PRODUCTS;
```

PRODUCT_NAME	PRICE	PRICE_CATEGORY
LAPTOP	1200,00	EXPENSIVE
PHONE	800,00	MID-RANGE
KEYBOARD	45,00	BUDGET
MONITOR	300,00	MID-RANGE
MOUSE	25,00	BUDGET

(2)

TABLE: ORDERS

ORDER_ID	CUSTOMER_NAME	AMOUNT
1	ALICE	150,00
2	BOB	560,00
3	CHARLIE	999,99
4	DIANA	45,50
5	ETHAN	1200,00

QUESTION: LABEL EACH ORDER

- 'HIGH VALUE' FOR ORDERS ≥ 1000
- 'MEDIUM VALUE' FOR 500 - 999,99
- 'LOW VALUE' FOR ORDERS < 500

EXPECTED OUTPUT COLUMNS

- CUSTOM_NAME
- AMOUNT
- ORDER_VALUE_CATEGORY

```
SELECT CUSTOMER_NAME, AMOUNT
CASE
    WHEN AMOUNT >= 1000 THEN 'HIGH_VALUE'
    WHEN AMOUNT BETWEEN 500 AND 999,99
ELSE 'LOW VALUE'
END AS ORDER_VALUE_CATEGORY
FROM ORDERS
```

CUSTOMER_NAME	AMOUNT	ORDER_VALUE_CATEGORY
ALICE	150,00	LOW VALUE
BOB	560,00	MEDIUM_VALUE
CHARLIE	999,99	MEDIUM_VALUE
DIANA	45,50	LOW VALUE
ETHAN	1200,00	HIGH VALUE

③ TABLE: EMPLOYEES

EMP_ID	EMP_NAME	DEPARTMENT	SALARY
1	JOHN	IT	85 000
2	SARA	HR	60 000
3	MARK	IT	75 000
4	LUCY	FINANCE	95 000
5	TOM	HR	55000

QUESTION: CATEGORIZE EMPLOYEE POSITION

- IF IN 'IT' AND SALARY > 80000 → 'SENIOR IT'
- IF IN 'HR' AND SALARY > 55000 → 'EXPERIENCED HR'
- OTHERWISE → 'STAFF'

EXPECTED OUTPUT COLUMNS

- EMP_NAME
- DEPARTMENT
- SALARY
- POSITION LEVEL

```
SELECT 'EMP_NAME', 'DEPARTMENT', 'SALARY
CASE
    WHEN 'DEPARTMENT' = 'IT' AND SALARY > 80000 THEN 'SENIOR IT'
    WHEN 'DEPARTMENT' = 'HR' AND SALARY > 55000 THEN 'EXPERIENCED HR'
    ELSE AS POSITION_LEVEL
FROM EMPLOYEES;
```

EMP_NAME	DEPARTMENT	SALARY	POSITION_LEVEL
JOHN	IT	85 000	SENIOR IT
SARA	HR	60 000	EXPERIENCED HR
MARK	IT	75000	STAFF
LUCY	FINANCE	95000	STAFF
TOM	HR	55000	STAFF

④ TABLE: STUDENTS

STUDENT_ID	STUDENT_NAME	SCORE
1	ANNA	92
2	BEN	76
3	CARA	59
4	DAVID	83
5	ELLA	68

QUESTION: ASSIGN A LETTER GRADE

- ≥ 90 : 'A'
- 80-89 : 'B'
- 70-79 : 'C'
- 60-69 : 'D'
- 60 < : 'F'

EXPECTED OUTPUT COLUMNS:

- STUDENT_NAME
- SCORE
- GRADE

```
SELECT STUDENT_NAME, SCORE
 CASE
     WHEN 'SCORE' >= 90 THEN 'A'
     WHEN 'SCORE' BETWEEN 80 AND 89 THEN 'B'
     WHEN 'SCORE' BETWEEN 70 AND 79 THEN 'C'
     WHEN 'SCORE' BETWEEN 60 AND 69 THEN 'D'
     ELSE '7'
 END AS GRADE
 FROM STUDENTS
```

STUDENT_NAME	SCORE	GRADE
ANNA	92	A
BEN	76	C
CARA	59	7
DAVID	83	B
ELLA	68	D

⑤ TABLE : DELIVERIES

DELIVERY_ID	DELIVERY_TIME_MINUTES
1	45
2	80
3	30
4	65
5	100

QUESTION : LABEL DELIVERY PERFOMANCE

- ≤ 30 MINS : 'FAST'
- 31-60 MINS : 'ON TIME'
- 60 MINS : 'LATE'

EXPECTED OUTPUT COLUMNS :

- DELIVERY_ID
- DELIVERY_TIME_MINUTES
- PERFOMANCE

```
SELECT DELIVERY_ID, DELIVERY_TIME_MINUTES PERFOMANCE
CASE
    WHEN DELIVERY_TIME_MINUTES < 30 THEN 'FAST'
    WHEN DELIVERY_TIME_MINUTES BETWEEN 31- 60 THEN 'ON TIME'
  ELSE 'LATE'
END AS PERFOMANCE
FROM DELIVERY
```

DELIVERY_ID	DELIVERY_TIME_MINUTES	PERFORMANCE
1	45	ON TIME
2	80	LATE
3	30	FAST
4	65	LATE
5	100	LATE

(6) TABLE: TICKETS

TICKET_ID	ISSUE_TYPE	PRIORITY
1	LOGIN ISSUE	1
2	SERVER DOWN	3
3	SLOW SYSTEM	2
4	EMAIL ERROR	2
5	PASSWORD RESET	1

QUESTION: CONVERT PRIORITY TO LABELS

- 3 → 'HIGH'
- 2 → 'MEDIUM'
- 1 → 'LOW'

EXPECTED OUTPUT COLUMNS

- ISSUE_TYPE
- PRIORITY
- PRIORITY_LEVEL

```
SELECT ISSUE_TYPE, PRIORITY
CASE
    WHEN PRIORITY = 3 THEN 'HIGH'
    WHEN PRIORITY = 2 THEN 'MEDIUM'
    ELSE 'LOW'
END AS PRIORITY_LABEL
FROM TICKETS
```

ISSUE TYPE	PRIORITY	PRIORITY_LABEL
LOGIN ISSUE	1	LOW
SERVE DOWN	3	HIGH
SLOW SYSTEM	2	MEDIUM
EMAIL ERROR	2	MEDIUM
PASSWORD RESET	1	LOW

7) TABLE: ATTENDANCE

STUDENT_ID	DAYS_PRESENT	TOTAL_DAYS
1	45	50
2	30	50
3	48	50
4	25	50
5	50	50

QUESTION: CALCULATE ATTENDANCE % CLASSIFY

- ≥ 90 % → 'EXCELLENT'
- 75-89 % → 'GOOD'
- < 75 % → 'NEEDS IMPROVEMENT'

EXPECTED OUTPUT COLUMNS

- STUDENT_ID
- ATTENDANCE_PERCENTAGE
- ATTENDANCE_STATUS

```
SELECT 'STUDENT_ID', 'TOTAL_DAYS', 'DAYS_PRESENT
('DAYS_PRESENT'/'TOTAL_DAYS') 100 AS ATTENDANCE_PERCENTAGE,
    CASE
        WHEN ('DAYS_PRESENT'/'TOTAL_DAYS') = 100 ≥ THEN 'EXCELLENT'
        WHEN ('DAYS-PRESENT'/'TOTAL_DAYS') = 100 BETWEEN 75 AND 89 THEN 'GOOD'
    ELSE 'NEEDS IMPROVEMENT'
  END AS ATTENDANCE_STATUS
FROM ATTENDANCE
```

STUDENT_ID	TOTAL_DAYS	DAYS_PRESENT	ATTEDANCE_PERCENTAGE	ATTEDANCE_STATUS
1	50	45	90	EXCELLENCE
2	50	30	60	NEED IMPROVE
3	50	48	96	EXCELLENCE
4	50	25	50	NEEDS IMPROVE.
5	50	50	100	EXCELLENCE

(8) TABLE : PRODUCTS_INVENTORY

PRODUCT_ID	STOCK_QTY
1	5
2	0
3	25
4	10
5	3

QUESTION : LABEL STOCK STATUS

- 0 → 'OUT OF STOCK'
- 1–5 → 'LOW STOCK'
- 5 → 'IN STOCK'

EXPECTED OUTPUT COLUMNS

- PRODUCT_ID
- STOCK_QTY
- STOCK_STATUS

```
SELECT 'PRODUCT_ID', 'STOCK_QTY', 'STOCK_STATUS',
    CASE
        WHEN 'STOCK_QTY' = 0 THEN 'OUT OF STOCK'
        WHEN 'STOCK_QTY' BETWEEN 1 AND 5 THEN 'LOW STOCK'
    ELSE 'IN STOCK'
END AS STOCK_STATUS
FROM PRODUCTS_INVENTORY
```

PRODUCT_ID	STOCK_QTY	STOCK_STATUS
1	5	LOW STOCK
2	0	OUT OF STOCK
3	25	IN STOCK
4	10	IN STOCK
5	3	LOW STOCK

⑨ TABLE: CLASSES

CLASS_ID	SUBJECT	ENROLLED_STUDENTS
1	MATH	30
2	ENGLISH	25
3	SCIENCE	15
4	ART	5
5	HISTORY	20

QUESTION: CLASSIFY BY SIZE

- $\geq 25 \rightarrow$ 'LARGE'
- $10-24 \rightarrow$ 'MEDIUM'
- $< 10 \rightarrow$ 'SMALL'

EXPECTED OUTPUT COLUMS

- SUBJECT
- ENROLLED_STUDENTS
- CLASS_SIZE_CATEGORY

```
SELECT SUBJECT, ENROLLED_STUDENTS,
CASE
    WHEN 'ENROLLED_STUDENTS' >= 25 THEN 'LARGE'
    WHEN 'ENROLLED_STUDENTS' BETWEEN 10 AND 24 THEN 'MEDIUM'
    WHEN 'ENROLLED_STUDENTS < 10 THEN 'SMALL'
END AS CLASS_SIZE_CATEGORY
FROM CLASSES
```

SUBJECT	DELIVERY_TIME_MINUTES	PERFOMANCE
MATH	45	ON TIME
ENGLISH	80	LATE
SCIENCE	30	FAST
ART	65	LATE
HISTORY	100	LATE

(10) TABLE: PAYMENTS

PAYMENT_ID	AMOUNT	PAYMENT_METHOD
1	50,00	CARD
2	200,00	CASH
3	150,00	CARD
4	75,00	PAYPAL
5	300,00	CASH

QUESTION: APPLY DISCOUNT FLAG

- IF PAYMENT_METHOD = 'CASH' AND AMOUNT ≥ 200 → 'ELIGIBLE FOR DISCOUNT'
- OTHERWISE → 'NOT ELIGIBLE'

EXPECTED OUTPUT COLUMNS

- PAYMENT_ID
- PAYMENT_METHOD
- AMOUNT
- DISCOUNT_ELIGIBILITY

```
SELECT PAYMENTID, PAYMENT_METHOD, AMOUNT,
  CASE
    WHEN 'PAYMENT_METHOD' = 'CASH' AND 'AMOUNT' >= 200 THEN 'ELIGIBLE FOR DISCOUNT'
  ELSE 'NOT ELIGIBLE'
END AS DISCOUNT_ELIGIBLE
FROM PAYMENTS
```

PAYMENT_ID	AMOUNT	PAYMENT_METHOD	PAYMENT_METHOD
1	50	CARD	NOT ELIGIBLE
2	200	CARD	ELIGIBLE FOR DISCOUNT
3	150	CARD	NOT ELIGIBLE
4	75	PAYPAL	NOT ELIGIBLE
5	300	CASH	ELIGIBLE FOR DISCOUNT